



# Federico Favia

SENIOR COMPUTER VISION ENGINEER · ML/DL ENGINEER

Born: 04.11.1996, Mantova, Italy

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## Summary

Experienced Computer Vision and Deep Learning Engineer with strong expertise in image processing, deep neural networks, and CV pipelines, including cloud and edge deployments. Proficient in managing projects and mentoring AI/ML teams, coordinating development activities, and liaising with customers to deliver scalable solutions. Specialized in designing object detection, recognition, tracking, and segmentation Production-ready models with end-to-end experience in data collection, annotation, model optimization and deployment.

## Work Experience

### Technology Data Reply IT - AI & CV BU

Milan, Italy - Hybrid

SENIOR DATA SCIENTIST CONSULTANT

May 2021 - PRESENT

- Assumed Technical Lead role of newly born CV Business Unit; deployed and scaled new WEB UI for final product of AI Turnaround Management System for flights at one of the biggest Italian airports, reaching 85% F1-score on multiple locations. Production code deployed on Kubernetes on-prem, refactored and DL Model fine-tuned with MLOps practices. Scaled it on cloud AKS for a another Italian airport.
- Optimized processes through computer vision and AI on network cameras for important airports customers (biggest European cargo airport which lowered its CO2 footprint and ground operation times), fine-tuning models reaching 82% mAP.
- Specialized in AI-driven solutions for smart operations, with deep expertise in computer vision, image processing, deep learning, and MLOps (Python, OpenCV, PyTorch, YOLOv5-v8, RTMDet, OCR, Azure ML Studio).
- Led small teams in end-to-end machine learning system development, from data collection, analysis and model training to production deployment, ensuring scalable and efficient solutions. Deployed object detection models and image processing pipelines on cloud platforms (Azure) and edge devices (NVIDIA Jetson).
- Designed and developed edge-based computer vision pipelines for airport security applications, achieving a 0.2% FalsePositive rate and enabling scalable deployment across multiple areas. Additionally built CV solutions for capacity monitoring and PRM (People with Reduced Mobility) detection to support operational efficiency improvements.
- Successfully supervised and executed full migration of several production-grade Azure resources, including AKS CV pipeline from company tenant to client tenant, ensuring full restoration of operational workflows.

### ManoMotion AB

Stockholm, Sweden

COMPUTER VISION ENGINEER INTERN

Jan. - Jul. 2020

- Deep Learning research project about hand segmentation to improve real-time AR hand tracking technology.
- Implementation and evaluation of different architectures in Python, Keras and Tensorflow: reached 96.03% Dice Accuracy on proprietary dataset and 75.26% Dice on average cross-dataset, running at 23.8 FPS (12.5 FPS on mobile device).

### Xtensa SRL

Trento, Italy

COMPUTER VISION STUDENT INTERN

Apr. - Jul. 2018

- Developed and validated a soccer player detection and tracking system using C/C++, OpenCV, and a convolutional neural network approach, specifically YOLOv3 on Darknet.

## Skills

|                  |                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>CS</b>        | Proficient in Python and OpenCV. Familiar in C/C++, Matlab, R and Arduino IDE, SQL, Snowflake, Postgres SQL                     |
| <b>AI</b>        | Proficient in PyTorch and YOLO, RTMDet Models, Cloud Deployment, Nvidia Jetson Edge Deployment, TensorFlow, Keras               |
| <b>MLOps</b>     | Proficient in MLOps and Azure Data Labeling, Pipelines in Azure ML Studio and Cloud Architectures, OCI Data Science             |
| <b>DevOps</b>    | Proficient in Docker, Kubernetes (K8S on-prem and AKS), DL Architecture Design, Bicep, Triton, TensorRT                         |
| <b>Back-end</b>  | MQTT, Azure ServiceBus and Azure EventHub. Familiar in Azure DataLake, Azure Data Factory, RabbitMQ                             |
| <b>Languages</b> | English C1 (IELTS Band 8), Italian                                                                                              |
| <b>Personal</b>  | Team Leading, Strong team-working and problem solving inclination, Mentoring and Training, Customer Communication               |
| <b>Others</b>    | Fluent in Bash, Git, Atlassian, Jira, Confluence, Microsoft Office and familiar with Adobe Illustrator/Photoshop; Scrum & Agile |

## Education

### KTH Royal Institute of Technology - Grade: A

Stockholm, Sweden

DOUBLE DEGREE MSc, ELECTRICAL ENGINEERING - ICT INNOVATION

2018 - 2021

- Final thesis project at company "Real-time hand segmentation using deep learning" contributing to new AR applications.
- Exit University of EIT Digital, Specialisation in Mobile Visual Computing: Courses in Computer Vision, Visual Data Analysis, Video Processing.

### University of Trento - Grade: 110/110 with honours

Trento, Italy

DOUBLE DEGREE MSc, INFORMATION AND COMMUNICATIONS ENGINEERING - ICT INNOVATION

2018 - 2021

- Entry University of EIT Digital Academy in Visual Computing Communication: courses in Pattern Recognition, Computer Graphics, Multimedia Security.
- Minor in Innovation and Entrepreneurship, School of Innovation CLAB Certificate.

### University of Trento - Grade: 105/110

Trento, Italy

BSc IN TELECOMMUNICATIONS ENGINEERING

2015 - 2018

- Final thesis "Convolutional Neural Networks' use for detection and tracking of players in sport events".
- Relevant courses in Image Processing and Transmission, Biomedical Diagnostic Techniques, Remote Sensing and ELEDIA ETRP Talented for Electromagnetic Field Simulators".

### Scientific High School - Grade: 100/100

Mantova, Italy

LSSA "E. FERMI"

2010 - 2015

## Projects

### Winners "Hack Future of SME Banking"

Budapest, Hungary

DEEPHACK EIT

Dec 2019

- Ultrahack & EIT FinTech Hackathon won with my team to improve OTP bank's E-biz platform.

### Startup Lab Trento

CLAB Trento, Italy

BUSINESS PROJECT

Feb 2019 - May 2019

- Business Development Laboratory: our team arrived in the finals with our IOT safety device for mountain bikers (awarded with best IoT project and best entrepreneurial project of the year).

### Computer Vision Academic Projects

Stockholm and Trento

C++, PYTHON, OPENCV, MATLAB

2018-2020

- Naval Traffic's Estimate in Venice, SIFT/SURF features for Image matching, Vocabulary Tree for Visual Search System, Edge Detection, Hough Transform and Object Segmentation, Image Compression.

## Certificates

|   |                                                                                                                                                                      |             |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| * | <b>Microsoft Azure: Data Scientist Ass. DP-100, AI Engineer Ass. AI-102, AI Fund. AI-900,</b>                                                                        | 2026 - 2023 |
| * | <b>M2L Mediterranean ML Summer School,</b> Milan Bicocca University                                                                                                  | Sep 2024    |
| * | <b>State Exam Qualification as Engineer - Informatics A,</b> Milan Polytechnic University                                                                            | Jan 2023    |
| * | <b>World Summit AI 2023,</b> Amsterdam                                                                                                                               | Oct 2023    |
| * | <b>IELTS Band 8 - C1 CEFR Level - General Training,</b> British Council                                                                                              | Jul 2022    |
| * | <b>OCI Data Science and DevOps Professional, OCI Machine Learning using Au- tonomous Database and Data Management, OCI Foundations, OCI AI Founda- tions,</b> Oracle | 2022-2023   |
| * | <b>NVIDIA - Fundamentals of Deep Learning,</b> NVIDIA DLI                                                                                                            | Sep 2021    |
| * | <b>Innovation Certificate,</b> School of Innovation Trento                                                                                                           | Aug 2021    |
| * | <b>Huawei Seeds for the Future,</b> Top-10 Tech Uni talents by Huawei Italy                                                                                          | Nov 2020    |
| * | <b>Helsinki EIT Summer School,</b> Aalto University, Disrupting Retail, Finland                                                                                      | Aug 2019    |
| * | <b>McKinksey Forward Program,</b> McKinsey                                                                                                                           | June 2026   |

## Awards

|      |                                                                                                                                   |       |
|------|-----------------------------------------------------------------------------------------------------------------------------------|-------|
| 2025 | <b>Reply Xchange Demo,</b> Computer Vision for Safety & Quality                                                                   | Milan |
| 2023 | <b>Reply Gala Awards,</b> Best Project Technology Innovation: "Advanced Trolley Management in Airport Ground Handling Operations" | Turin |
| 2023 | <b>Reply Xchange Speaker,</b> A Computer Vision AI Powered Solution for Smart Operation Control                                   | Milan |
| 2022 | <b>Reply Xchange Speaker,</b> Airport Intelligence: the Fraport Airfield Operations experience                                    | Milan |